

# Substitution $\phi[x/t]$

## Example (of substitution)

$$P(x, y, f(x))[x/a] = P(a, y, f(a))$$

$$\forall x P(x, y)[x/b] = \forall x P(x, y)$$

$$\exists x P(x, x) \wedge Q(x)[x/c] = \exists x P(x, x) \wedge Q(c)$$

$$P(x, g(y))[y/f(x)] = P(x, g(f(x)))$$

$$\forall x. P(x, y)[y/f(x)] =$$

Not allowed since  $f(x)$  is  
not free for  $y$  in  $\forall x. P(x, y)$

# Universal quantification rule

$$\frac{\forall x \phi(x)}{\phi(t)}$$

- $\forall x \phi(x)$  means that for every object of the domain, the property  $\phi(x)$  should be true.
- a term  $t$  that occurs in the tableaux denotes an object of the domain
- therefore,  $\phi(t)$  must be true for all the terms  $t$  that occurs in the tableaux. I.e., the  $\forall$  rule can be applied as many time as one want to any term that appear in the tableaux.

# Existential quantification rule

$$\frac{\exists x\phi(x)}{\phi(c)} \text{ for a new constant } c$$

- $\exists x\phi(x)$  means that **for some object** of the domain, the property  $\phi(x)$  should be true.
- we **don't know which object of the domain** has the property  $\phi$ , we know only that there is one.
- this means that this rule cannot be applied to the terms that already occur in the tableaux, since otherwise we would introduce an unjustified joiche on the element that has the property  $\phi$ .
- the trick is to introduce a term to denote an unconditioned objects (sometimes called “fresh” constant/variable) for denoting an “unknown” object, i.e., an object on which we haven't done any commitment.
- therefore we allow only to infer  $\phi(c)$  from  $\exists x\phi(x)$ , where  $c$  is fresh. Only one application is possible.

# Open and Closed Branches

- a tableaux rooted with  $\phi$  is a method to search an interpretation that satisfy  $\phi$
- Every branch of a tableaux with root equal to  $\phi$ , corresponds to an attempt to find an interpretation  $\mathcal{I}$  that satisfies  $\phi$ .
- The interpretation corresponding to a branch  $b$  of a tableaux should satisfy all the formulas that appear in the branch.
- If the branch contains two opposite literals, i.e.  $P(t_1, \dots, t_n)$  and  $\neg P(t_1, \dots, t_n)$ , then the branch cannot correspond to an interpretation, since there is no interpretation that satisfy at the same time  $P(t_1, \dots, t_n)$  and  $\neg P(t_1, \dots, t_n)$ . So we can consider this attempt to find an interpretation failed. In this case we say that the branch is **closed**.
- if in a branch  $b$  all the rules has been applied and there is no opposite literals, then this branch corresponds to an interpretation. We call such a branch **open**

# Open and Closed Branches

## Definition

- A branch of a tableau is said to be **closed** if it contains a pair of formulas  $\phi$  and  $\neg\phi$ .
- A branch of a tableau is said to be **open** if it is not closed.
- A tableau is said to be closed if each of its paths is closed.

# The tableaux method

- 1 To test a formula  $\phi$  for validity, form tableau starting with  $\neg\phi$ . If the tableau closes off, then  $\phi$  is logically valid.
- 2 To test whether  $\phi$  is a logical consequence of  $\Gamma$  form a tableau starting with each formula in  $\Gamma$  and  $\neg\phi$ . If the tableau closes off, then  $\phi$  is indeed a logical consequence of  $\Gamma$ .
- 3 To test a set of formulas  $\Gamma$  is satisfiable, form a tableau starting with  $\Gamma$  or equivalently an unsigned If the tableau closes off, then  $\Gamma$  is not satisfiable. If the tableau does not close off, then  $\Gamma$  is satisfiable, and from any open branch we can read off an interpretation satisfying  $\Gamma$ .

# Example

## Example

To check if the formula  $(\exists x(P(x) \vee Q(x))) \equiv ((\exists x P(x)) \vee (\exists x Q(x)))$  is ~~satisfiable~~ <sup>valid</sup>, we start with a tableaux with this formula:

$$\begin{array}{c}
 \neg((\exists x (Px \vee Qx)) \Leftrightarrow ((\exists x Px) \vee (\exists x Qx))) \\
 / \qquad \qquad \qquad \backslash \\
 \begin{array}{c}
 \exists x (Px \vee Qx) \\
 \neg((\exists x Px) \vee (\exists x Qx)) \\
 \neg \exists x Px \\
 \neg \exists x Qx \\
 Pa \vee Qa \\
 / \quad \backslash \\
 Pa \quad Qa \\
 \neg Pa \quad \neg Qa
 \end{array}
 \qquad
 \begin{array}{c}
 \neg \exists x (Px \vee Qx) \\
 (\exists x Px) \vee (\exists x Qx) \\
 / \qquad \qquad \backslash \\
 \begin{array}{c}
 \exists x Px \\
 Pb \\
 \neg(Pb \vee Qb) \\
 \neg Pb \\
 \neg Qb
 \end{array}
 \qquad
 \begin{array}{c}
 \exists x Qx \\
 Qc \\
 \neg(Pc \vee Qc) \\
 \neg Pc \\
 \neg Qc
 \end{array}
 \end{array}$$

## Exercise

Show with the method of semantic tableaux that the following formulas are valid:

- $\forall x P(x) \supset \neg \exists x \neg P(x)$
- $\forall x (P(x) \vee A) \supset (\forall x P(x) \vee A)$  when  $x$  is not free in  $A$
- $\exists x (P(x) \supset \forall x P(x))$
- $\exists x \forall y P(x, y) \supset \forall y \exists x P(x, y)$

# Practicing with Semantic Tableaux

## Solution

$$\neg(\forall x P(x) \supset \neg \exists x \neg P(x))$$

$$\begin{array}{c} \forall x P(x) \\ \neg \neg \exists x \neg P(x) \end{array}$$

$$\exists x \neg P(x)$$

$$\neg P(a)$$

$$P(a)$$

x

$$\neg(\forall x(P(x) \vee A) \supset (\forall x P(x) \vee A))$$

$$\begin{array}{c} \forall x(P(x) \vee A) \\ \neg(\forall x P(x) \vee A) \end{array}$$

$$\neg \forall x P(x)$$

$$\neg A$$

$$\neg P(a)$$

$$P(a) \vee A$$

$$P(a)$$

$$A$$

x

x

# Practicing with Semantic Tableaux

## Solution

$$\neg \exists x (P(x) \supset \forall x P(x))$$

$$\neg (P(a) \supset \forall x P(x))$$

$$\begin{array}{c} P(a) \\ \neg \forall x P(x) \end{array}$$

$$\neg P(b)$$

$$\neg (P(b) \supset \forall x P(x))$$

$$\begin{array}{c} P(b) \\ \neg \forall x P(x) \end{array}$$

x

$$\neg (\exists x \forall y P(x, y) \supset \forall y \exists x P(x, y))$$

$$\begin{array}{c} \exists x \forall y P(x, y) \\ \neg \forall y \exists x P(x, y) \end{array}$$

$$\forall y P(a, y)$$

$$\neg \exists x P(x, b)$$

$$P(a, b)$$

$$\neg P(a, b)$$

x

# Example

## Example

Check if  $\forall x P(x) \wedge \exists x \neg P(f(x))$  is valid/satisfiable/unsatisfiable.

## Solution

$$\forall x P(x) \wedge \exists x \neg P(f(x))$$

|

$$\forall x P(x)$$

$$\exists x \neg P(f(x))$$

|

$$\neg P(f(c))$$

# Example

## Example

Check if  $\forall x P(x) \wedge \exists x \neg P(f(x))$  is valid/satisfiable/unsatisfiable.

## Solution

$$\forall x P(x) \wedge \exists x \neg P(f(x))$$

|

$$\forall x P(x)$$

$$\exists x \neg P(f(x))$$

|

$$\neg P(f(c))$$

*Now to expand  $\forall x P(x)$ , we can use any ground term  $t$ . possible choices:  $c$ ,  $f(c)$ ,  $f(f(c))$ , .... we choose  $f(c)$  because we want to create a clash with  $\neg P(f(c))$ .*

## Example (Cont'd)

### Example

Check if  $\forall x P(x) \wedge \exists x \neg P(f(x))$  is valid/satisfiable/unsatisfiable.

### Solution

$$\forall x P(x) \wedge \exists x \neg P(f(x))$$

|

$$\forall x P(x)$$

$$\exists x \neg P(f(x))$$

|

$$\neg P(f(c))$$

|

$$P(f(c))$$

|

x

# Example of tableaux

## Example

$$\exists x(P(x) \wedge \neg Q(x)) \wedge \forall y(P(y) \vee Q(y))$$

$$\begin{array}{c} | \\ \exists x(P(x) \wedge \neg Q(x)) \\ \forall y(P(y) \vee Q(y)) \end{array}$$

$$\begin{array}{c} | \\ P(a) \wedge \neg Q(a) \end{array}$$

$$\begin{array}{c} | \\ P(a) \\ \neg Q(a) \end{array}$$

$$\begin{array}{c} | \\ P(a) \vee Q(a) \end{array}$$

$$\begin{array}{c} / \\ P(a) \end{array}$$

$$\begin{array}{c} \backslash \\ Q(a) \end{array}$$

OPEN

CLASH

# Termination

$$\exists x.P(x) \wedge \forall x(P(x) \supset P(f(x)))$$

$$\begin{array}{c} \exists x.P(x) \\ \forall x(P(x) \supset P(f(x))) \end{array}$$

$$P(a)$$

$$P(a) \supset P(f(a))$$

$$\neg P(a)$$

CLASH

$$P(f(a))$$

$$P(f(a)) \supset P(f(f(a)))$$

$$\neg P(f(a))$$

CLASH

$$P(f(f(a)))$$

⋮

For certain formulas  
the tableaux can go on  
for ever generating an  
infinite tree. Consider,  
for instance the  
tableaux for the  
formula  
 $\exists x.P(x) \wedge (\forall xP(x) \supset P(f(x)))$